

Group Presentation and Evaluation 2

This group assignment combines all topics covered up to and including Week 12. Submit your completed R Markdown file (RMD) and either a copy of your presentation slides (presenters) or completed evaluation rubric (evaluators) onto Moodle.

Note the following:

- This is a ‘Guided Gen AI Use’ assignment where you are permitted to use generative AI (GenAI) as a supporting tool. Following University guidelines, if GenAI is used, you must submit a short declaration describing the tools used, purpose of use, extent of use, and outline the steps taken to verify the accuracy of the GenAI output.
- You cannot manually modify the data in software like Excel. Anything related to data preparation and analysis must be completed using R.
- You are expected to utilise the techniques covered in this course, where appropriate, and are not required to go beyond them.
- Maintain a tidy and readable RMD file that runs without any errors. Marks will be deducted if reported findings cannot be replicated from running the submitted script.

Data:

The data for this assignment were extracted from Tripadvisor. It consists of information on hotels in the Waikato Region and customer reviews posted before the end of July 2026 for selected hotels in Hamilton and nearby cities.

week14_hotelmain.csv, *week14_hotelpred.csv* and *week14_review.csv* include the data to use for this assignment. Refer to later pages for variable names and definitions.

Scenario:

Your client, the manager of Ramada by Wyndham Hamilton City Center (*id: 3572177*), wants to identify strengths and areas for improvement in the hotel’s operations and understand how it compares with its competitors. The manager is also considering increasing their price per night, which currently averages \$183.50 per night when incorporating discounts offered. Prior to implementing any pricing adjustments, the manager wants an estimate of what their hotel should be priced at based on a machine learning model.

You have extracted reviews on the client’s hotel and hotels nearby from a travel website. You have also gathered data on hotels in the Waikato region to build a prediction model. Conduct your analysis using R and report your preliminary findings to the manager.

Requirements:

1. Conduct appropriate and comprehensive analysis to provide insights on the hotel's strengths and areas for improvement and how it compares with competitors.
2. Build the best model you can for predicting average price per night (*price*) and evaluate model performance.
3. Summarise your main findings and present data-informed recommendations on how the manager can improve their operations and whether the current price per night should be adjusted.

Variable names and definitions:

Dataset: *week14_hotel*.csv*

Variable name	Description
id	Tripadvisor unique hotel ID
name	Hotel name
description	Hotel description
email	Hotel email contact
website	Hotel website
phone	Hotel phone number
category	Type of accommodation
address	Hotel address
country	Country the hotel is located
locality	City or municipal the hotel is located
postalcode	Hotel postal code
latitude	Latitude coordinate
longitude	Longitude coordinate
hotel_class	Rating out of 5 stars by an independent hotel mapping organisation. Indicates the general level of features, amenities, and services to expect
price_class	The relative cost level of a hotel based on Tripadvisor's price classification, ranging from \$ (budget) to \$\$\$\$ (high-end)
price	Average price per night for a standard room
room	Total number of rooms
photo	Total number of photos available of the hotel on Tripadvisor
offer_vendors	Number of booking providers associated with the hotel (e.g., Booking.com, Expedia, direct hotel website)
offer_pricemin	Lowest advertised price for the hotel across all booking providers
offer_pricemax	Highest advertised price for the hotel across all booking providers
amenity_airport	Hotel provides airport transportation services
amenity_baggage	Hotel provides baggage storage services
amenity_breakfastava	Hotel provides complimentary access to breakfast
amenity_breakfastinc	Hotel provides breakfast with the booking
amenity_housekeep	Hotel provides housekeeping services
amenity_kids	Hotel provides activities or play zones for children
amenity_kitchen	Hotel provides kitchenette
amenity_laundryself	Hotel provides self-serve laundry facilities
amenity_laundryserv	Hotel provides laundry services

amenity_nonsmoke	Hotel provides non-smoking rooms
amenity_parking	Hotel provides free parking
amenity_pool	Hotel provides a swimming pool
amenity_service	Hotel provides room service
amenity_wheelchair	Hotel provides wheelchair access
amenity_wifi	Hotel provides free Wi-Fi
rank_string	Text displayed on Tripadvisor that describes the hotel's position within a destination and accommodation category
rank_position	Numerical rank of the hotel within its destination and accommodation category
rank_denominator	Total number of hotels in the same destination and accommodation ranking category
review	Total number of reviews received
rating_overall	Average reviewer rating out of 5 for the hotel overall
rating_cleanliness	Average reviewer rating out of 5 for cleanliness and hygiene
rating_location	Average reviewer rating out of 5 for convenience and accessibility of the hotel
rating_room	Average reviewer rating out of 5 for the hotel room
rating_service	Average reviewer rating out of 5 for service quality
rating_sleep	Average reviewer rating out of 5 for the quality of sleep experienced
rating_value	Average reviewer rating out of 5 for perceived value for money

Dataset: *week14_review.csv*

Variable name	Description
hotel_id	Tripadvisor unique hotel ID
review_id	Tripadvisor unique review ID
review_helpful	Total number of 'helpful' votes received by the review
user_id	Tripadvisor unique reviewer ID
user_location	Reviewer location as registered on Tripadvisor
user_review	Total number of reviews provided by the reviewer
travel_date	Date the reviewer visited the hotel
trip_type	Type of trip
review_date	Date the review was posted
title	Review title
text	Main review text
rating_overall	Reviewer rating out of 5 for the hotel overall
rating_cleanliness	Reviewer rating out of 5 for cleanliness and hygiene
rating_location	Reviewer rating out of 5 for convenience and accessibility of the hotel
rating_room	Reviewer rating out of 5 for the hotel room
rating_service	Reviewer rating out of 5 for service quality
rating_sleep	Reviewer rating out of 5 for the quality of sleep experienced
rating_value	Reviewer rating out of 5 for perceived value for money
hotel_response	Hotel response to the review